RAJEEV INSTITUTE OF TECHNOLOGY

HASSAN-573201



Internet of Things Lab

(BCSL701)

As per VTU Syllabus/scheme for 7th Semester

Faculty Name: Kusuma H M
Asst.professor

DEPARTMENT OF

COMPUTER SCIENCE & ENGINEERING

VISION & MISSION OF THE INSTITUTE

VISION:

- To be an academic institution in vibrant social & economic environment, striving continuously for excellence in education, research and technological service to the society.

MISSION:

- To achieve academic excellence in engineering and management through dedication to duty, offering state of the art education and faith in human values.
- To create and endure a community of learning among students, develop outstanding professionals with high ethical standards.
- To provide academic ambience conducive to the development, needs and growth of society and the industry.

VISION & MISSION OF THE DEPARTMENT

VISION:

- To become an exemplary center for engineering education and research in the frontier areas of Computer Science and Engineering.

MISSION:

- Render quality education through best teaching learning process and modern educational tools to enable students for good careers.
- Impart essential skills to make students industry ready and future professional with social concern.
- Provide facilities and expertise in recent computer technology to promote research and innovations.

PROGRAM SPECIFIC OUTCOMES (PSO'S)

The graduates of Computer Science & engineering program of Rajeev Institute of Technology should be able to attain the following at the time of graduation.

PSO1: To apply software engineering principles and practices to design, develop and implement software solutions.

PSO2: To use recent software tools to solve and analyze a given problem.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

The program educational objectives are the statements that describe the expected achievements of graduates within first few years of their graduation from the program. The program educational objectives of Bachelor of Computer Science & Engineering at Rajeev Institute of Technology can be broadly defined as,

PEO1: Graduates will become software developers in diverse domains and/or pursue higher studies.

PEO2: Graduates will process an agility to adapt to challenges in their profession.

PEO3: Graduates exhibit leadership qualities and will be a team player.

PEO4: Graduates will be able to communicate effectively and cultivate ethics.

PROGRAM OUTCOMES

Graduation students of Bachelor of Computer Science and Engineering program at Rajeev Institute of Technology will attain the following program outcomes in the field of Computer Science and Engineering.

PO1-Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2-Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3-Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4-Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5-Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling to complex engineering activities, with an understanding of the limitations.

PO6-The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7-Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8-Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9-Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10-Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11-Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environment.

PO12-Life-long learning: Recognize the need for, and have the preparation and ability to engage in in dependent and life-long learning in the broadest context of technological change.

EXPERIMENT INDEX

SL.NO	Experiment Name
1	**Develop a program to blink 5 LEDs back and forth.**
2	**Develop a program to interface a relay with Arduino board.**
3	**Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors**
4	**Develop a program to control a DC motor with Arduino board.**
5	**Develop a program to deploy smart street light system using LDR sensor.**
6	**Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).**
7	**Develop a program to read the pH value of various substances like milk, lime and water.**
8	**Develop a program to detect the gas leakage in the surrounding environment.**
9	**Develop a program to demonstrate weather station readings using Arduino.**
10	**Develop a program to setup a UART protocol and pass a string through the protocol.**
11	**Develop a water level depth detection system using Ultrasonic sensor.**
12	**Develop a program to simulate interfacing with the keypad module to record the keystrokes.**

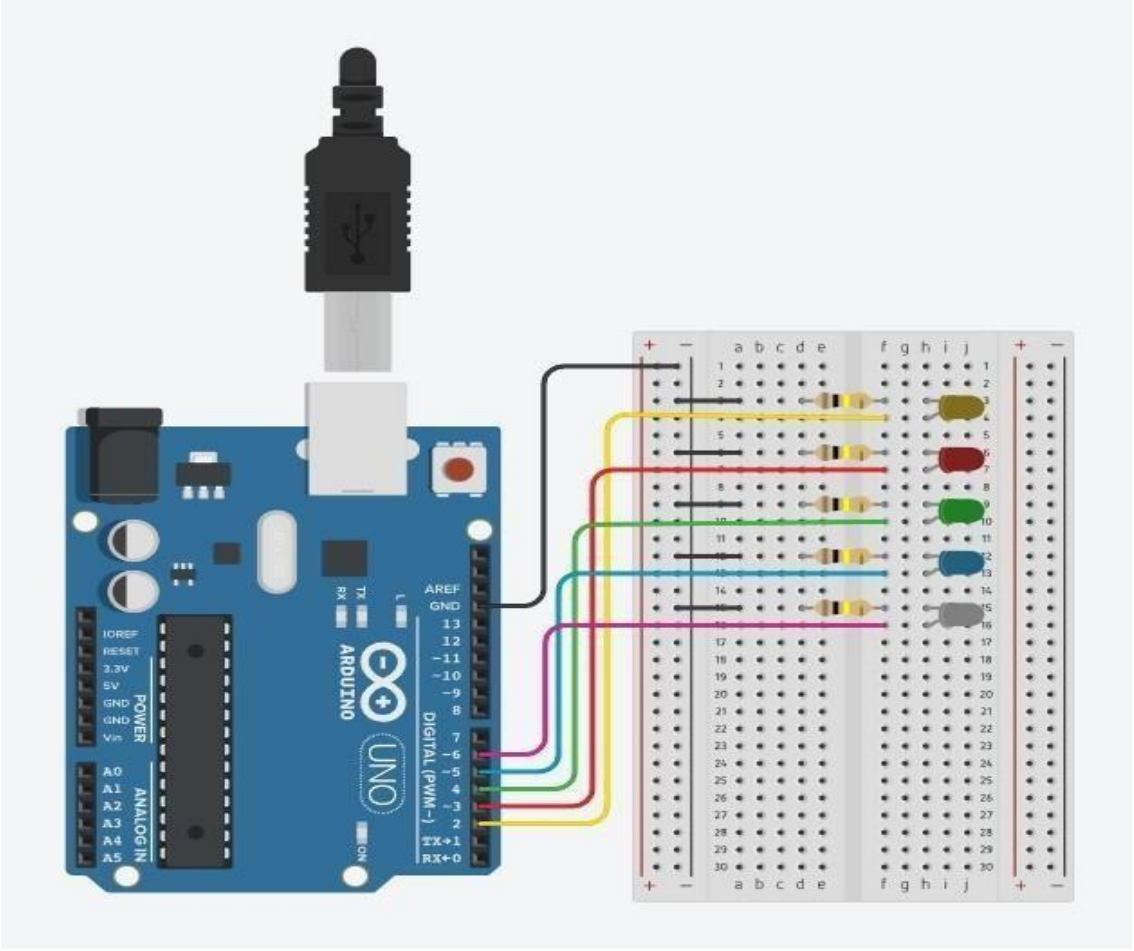
Experiment 1: Develop a program to blink 5 LEDs back and forth.

AIM: Develop a program to blink 5 LEDs back and forth.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**6**
3	Bread Board	**1**
4	LED	**5**
5	Resistance (800 Ω)	**5**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE, then go to Tools and select appropriate Arduino board.

d) Select Tools, then select the Port and choose the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
void setup() {
pinMode(2, OUTPUT);
pinMode(3, OUTPUT);
pinMode(4, OUTPUT);
pinMode(5, OUTPUT);
pinMode(6, OUTPUT);
}
void loop() {
digitalWrite(2, HIGH);
delay(1000);
digitalWrite(2, LOW);
delay(100);

digitalWrite(3, HIGH);
delay(1000);
digitalWrite(3, LOW);
delay(100);
```

```
digitalWrite(4, HIGH);
delay(1000);
digitalWrite(4, LOW);
delay(100);

digitalWrite(5, HIGH);
delay(1000);
digitalWrite(5, LOW);
delay(100);

digitalWrite(6, HIGH);
delay(1000);
digitalWrite(6, LOW);
delay(100);

digitalWrite(5, HIGH);
delay(1000);
digitalWrite(5, LOW);
delay(100);

digitalWrite(4, HIGH);
delay(1000);
digitalWrite(4, LOW);
delay(100);

digitalWrite(3, HIGH);
delay(1000);
```

```
digitalWrite(3, LOW);
delay(100);

digitalWrite(2, HIGH);
delay(1000);
}
```

RESULT : Successfully demonstrated blink 5 LEDs.

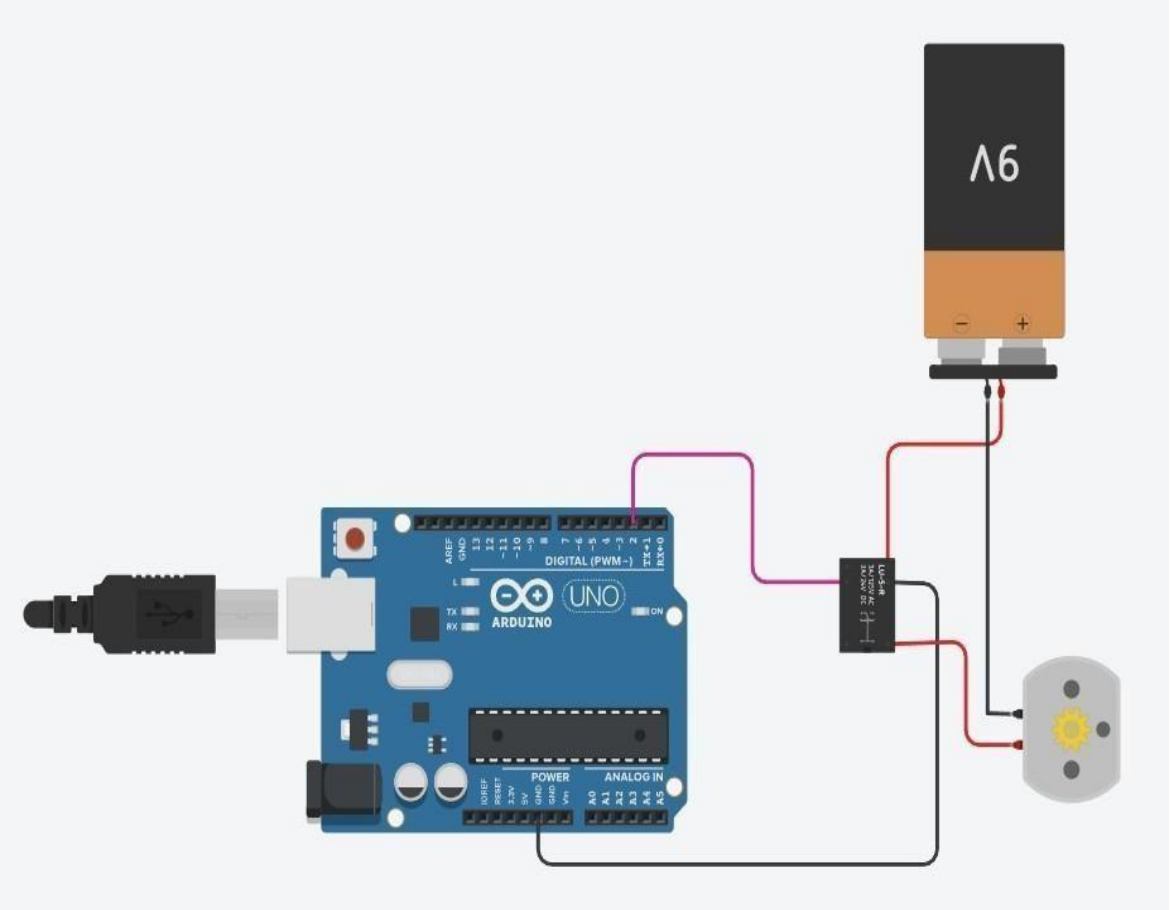
Experiment 2. Develop a program to interface a relay with Arduino board.

AIM: Develop a program to interface a relay with Arduino board.

COMPONENTS:

SL NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**6**
3	DC Motor	**1**
4	Relay SPDT	**1**
5	9V Battery	**1**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
void setup()
{
pinMode(2, OUTPUT);
}
void loop()
{
digitalWrite(2, HIGH);
delay(10000);
digitalWrite(2, LOW);
delay(5000);
}
```

RESULT: Successfully demonstrated interface a relay with Arduino board.

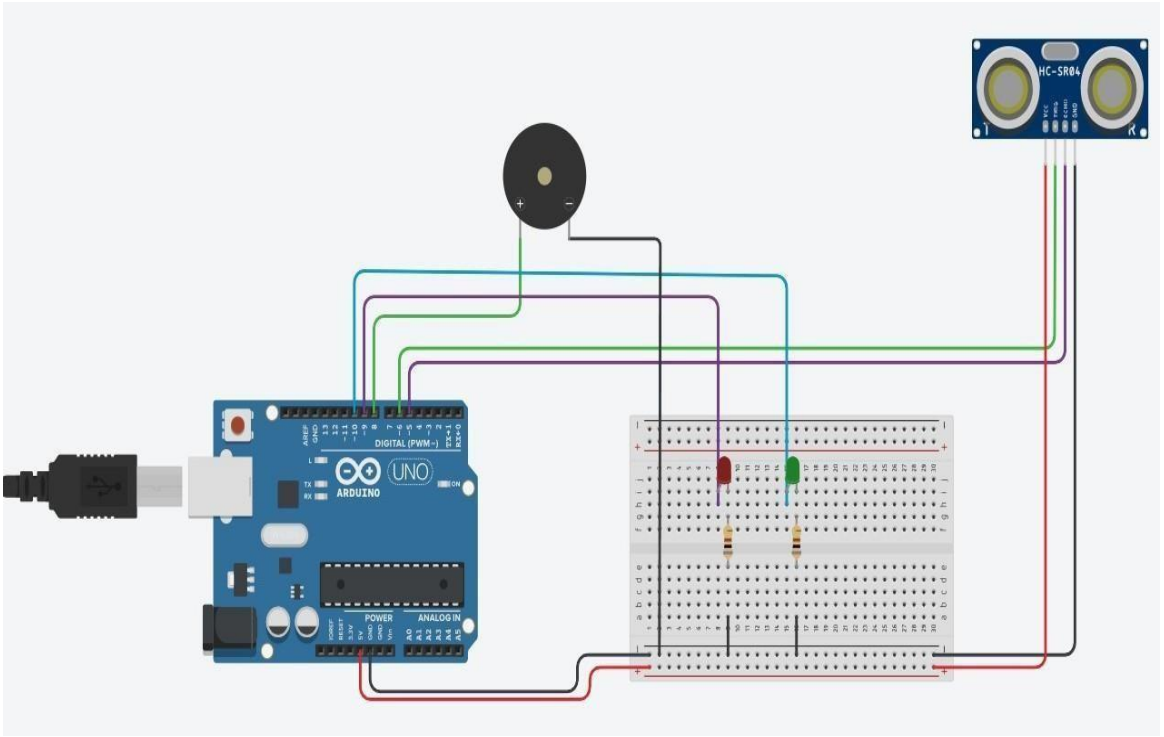
Experiment 3. Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

AIM: Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**6**
3	Bread Board	**1**
4	LED	**2**
5	Resistance (800 Ω)	**2**
6	Ultrasonic Distance Sensor (4-pin)	**1**
7	Piezo	**1**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

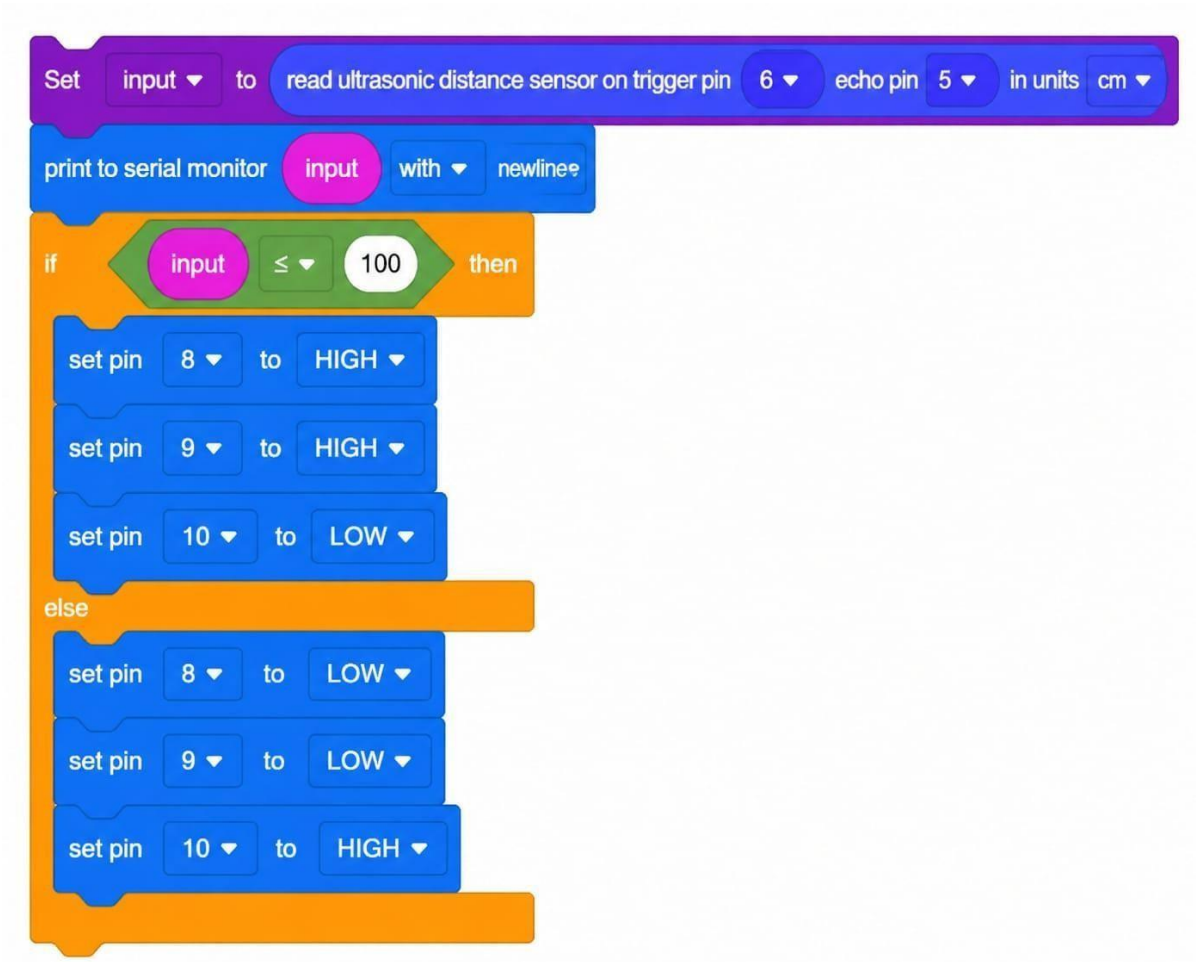
b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:



RESULT: Successfully demonstrated deploy an intrusion detection system using Ultrasonic and sound sensors.

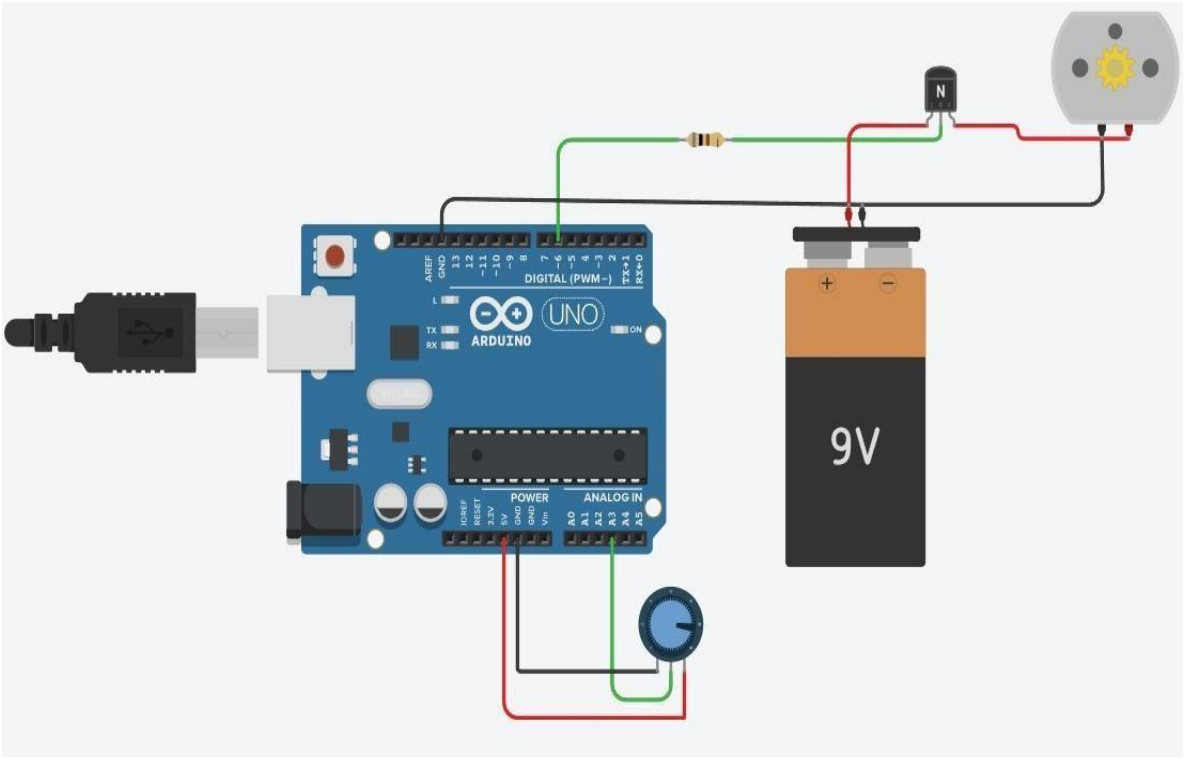
Experiment 4: Develop a program to control a DC motor with Arduino board

AIM: To develop a program to control a DC motor using an Arduino board.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	1
2	Jumper Cable	6
3	Resistance(800 Ω)	1
4	DC Motor	1
5	NPN transistor(BJT)	1
6	9V Battery	1
7	250 kΩ Potentiometer	1

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
const int poten = A3;
int var;
void setup()
{
  pinMode(6, OUTPUT);
}
void loop()
{
  var = analogRead(poten);
  int pwmValue = map(var, 0, 1023, 0, 255);
  analogWrite(6, pwmValue);
}
```

RESULT: Successfully demonstrated control of a DC motor with Arduino board.

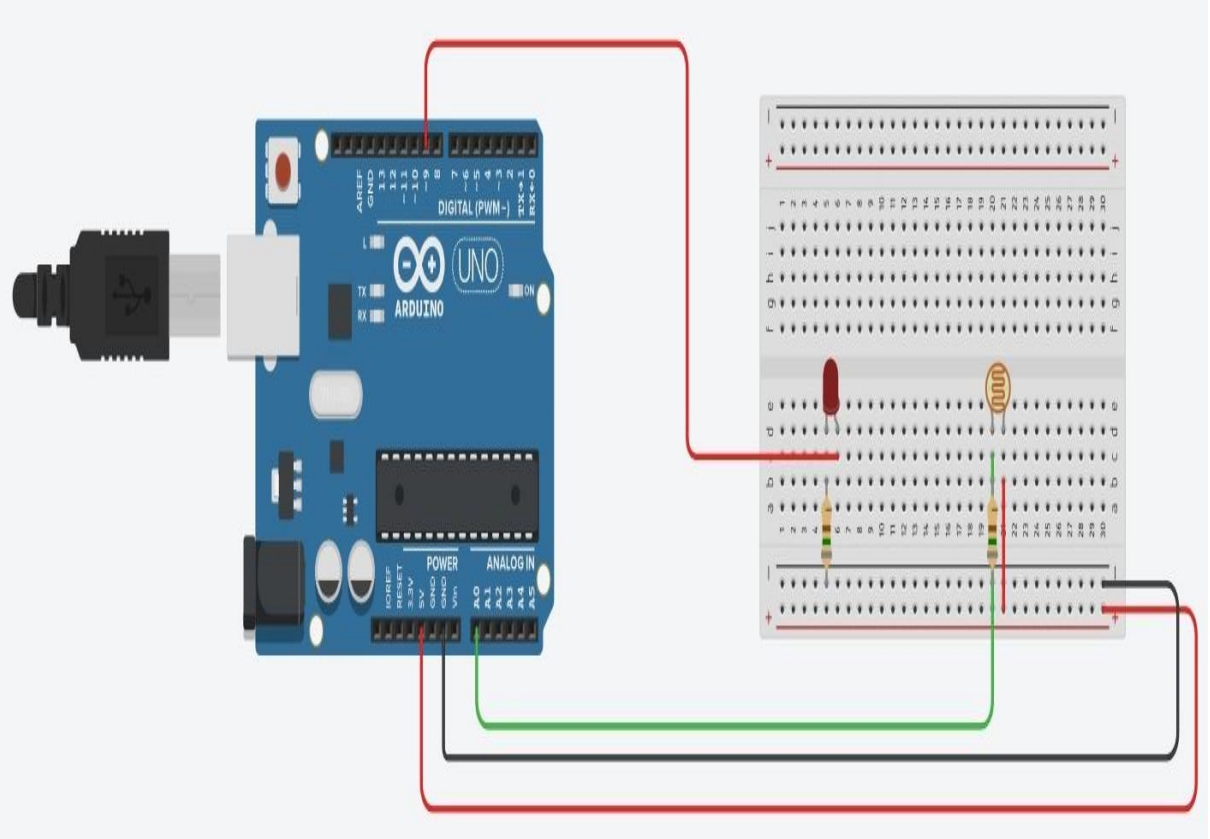
Experiment 5. Develop a program to deploy smart street light system using LDR sensor.

AIM: Develop a program to deploy smart street light system using LDR sensor.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**4**
3	Bread Board	**1**
4	LED	**1**
5	Resistance (800 Ω)	**2**
6	LDR	**1**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
int sensorPin = A0;
int sensorValue = 0;
int led = 9;
void setup()
{
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}
void loop()
 {
  sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  if(sensorValue < 100)
  {
    Serial.println("LED light on");
    digitalWrite(led, HIGH);
    delay(1000);
    digitalWrite(led, LOW);
  }
```

```
else
{
  digitalWrite(led, LOW);
}

delay(200);
}
```

RESULT: Successfully demonstrated deploy smart street light system using LDR sensor.

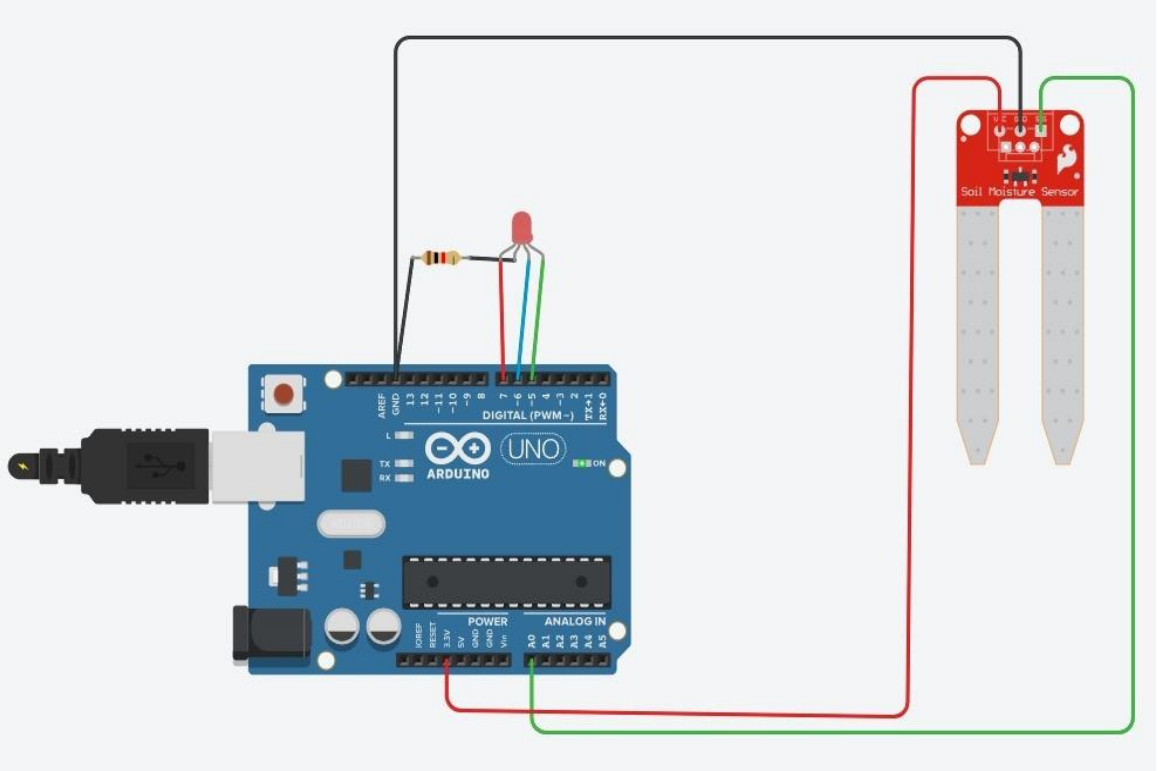
Experiment 6. Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

AIM: Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**8**
3	LED	**1**
5	Resistance (1KΩ)	**1**
6	Moisture Sensor (DHT22)	**1**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE

```
int moistureValue;
float moisture_percentage;
void setup()
{
  pinMode(7, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(5, OUTPUT);
  Serial.begin(9600);
}
void loop()
{
  moistureValue = analogRead(A0);
  moisture_percentage = (moistureValue * 100.0) / 539.0;
  if (moisture_percentage >= 0 && moisture_percentage < 25)
  {
    digitalWrite(7, HIGH);
    digitalWrite(6, LOW);
    digitalWrite(5, LOW);
  }
```

```
    else if (moisture_percentage >= 25 && moisture_percentage < 80)
    {
      digitalWrite(7, LOW);
      digitalWrite(6, HIGH);
      digitalWrite(5, LOW);
    }
    else if (moisture_percentage >= 80 && moisture_percentage <= 100)
    {
      digitalWrite(7, LOW);
      digitalWrite(6, LOW);
      digitalWrite(5, HIGH);
    }
    Serial.print("Moisture Value : ");
    Serial.print(moisture_percentage);
    Serial.println("%");
    delay(1000);
  }
```

RESULT: Successfully demonstrated dry and wet waste classification using the Moisture sensor (DHT22).

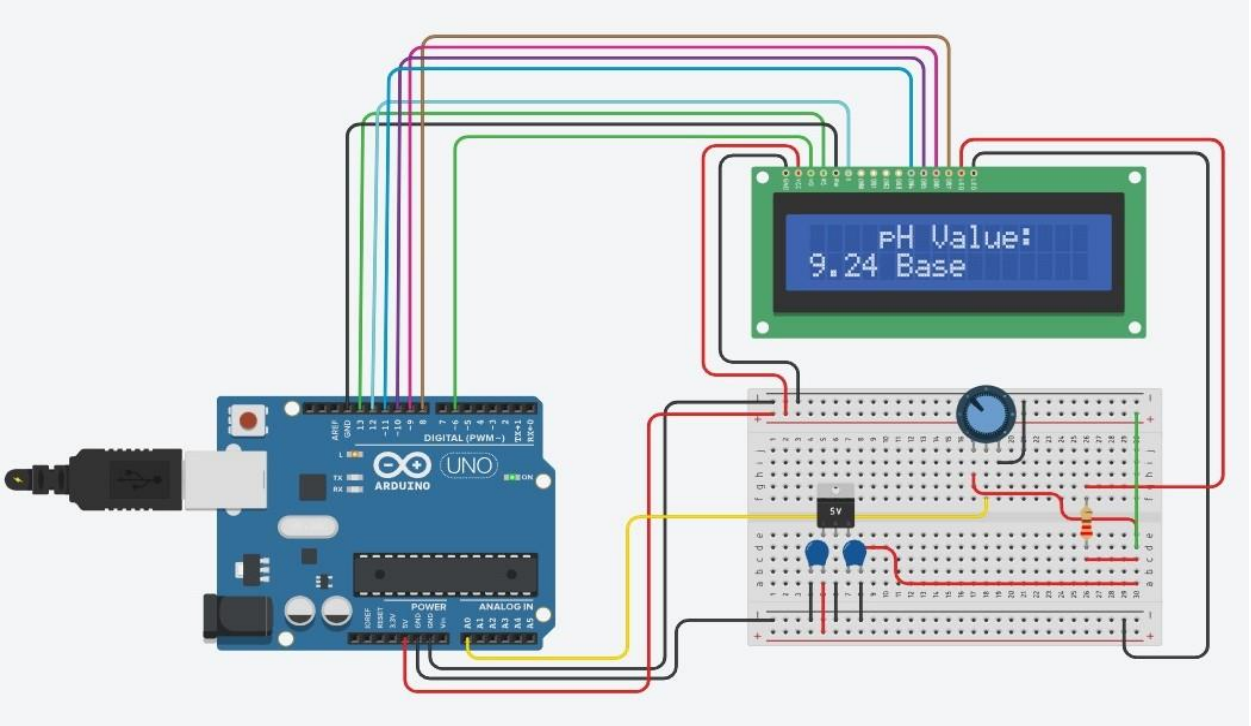
Experiment 7. Develop a program to read the pH value of various substances like milk, lime and water.

AIM: Develop a program to read the pH value of various substances like milk, lime and water.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**12**
3	Bread Board	**1**
4	Resistance (220Ω)	**1**
5	LCD 16 x 2	**1**
6	10 MΩ Potentiometer	**1**
7	5V Regulator (LM7805)	**1**
8	0.22µF Capacitor	**1**
9	0.1µF Capacitor	**1**

CIRCUIT DIAGRAM



SETUP :

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE

```
#include <LiquidCrystal.h>
const int rs = 13, en = 12, d4 = 11, d5 = 10, d6 = 9, d7 = 8;
 LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
 int Contrast = 0;
void setup()
{
  Serial.begin(9600);
 analogWrite(6, Contrast);
  lcd.begin(16, 2);
  lcd.setCursor(4, 0);
  lcd.print("pH Value:");
}
void loop()
{
  int sensorValue = analogRead(A0);
```

```
 float ph = sensorValue * (14.0 / 1023.0);
 Serial.println(ph);
 lcd.setCursor(0, 1);
 lcd.print("        ");
 lcd.setCursor(0, 1);
 lcd.print(ph, 2);
 lcd.print(" ");
 if (ph >= 0.0 && ph < 5.0)
 {
  lcd.print("ACID");
 }
 else if (ph >= 5.0 && ph < 7.0)
 {
  lcd.print("Normal");
 }
 else if (ph >= 7.0 && ph <= 14.0)
 {
  lcd.print("Base");
 }
 else
 {
  lcd.print("Invalid");
 }
```

```
  delay(1000);
}
```

RESULT : Successfully demonstrated read the pH value of a various substances like milk, lime and water.

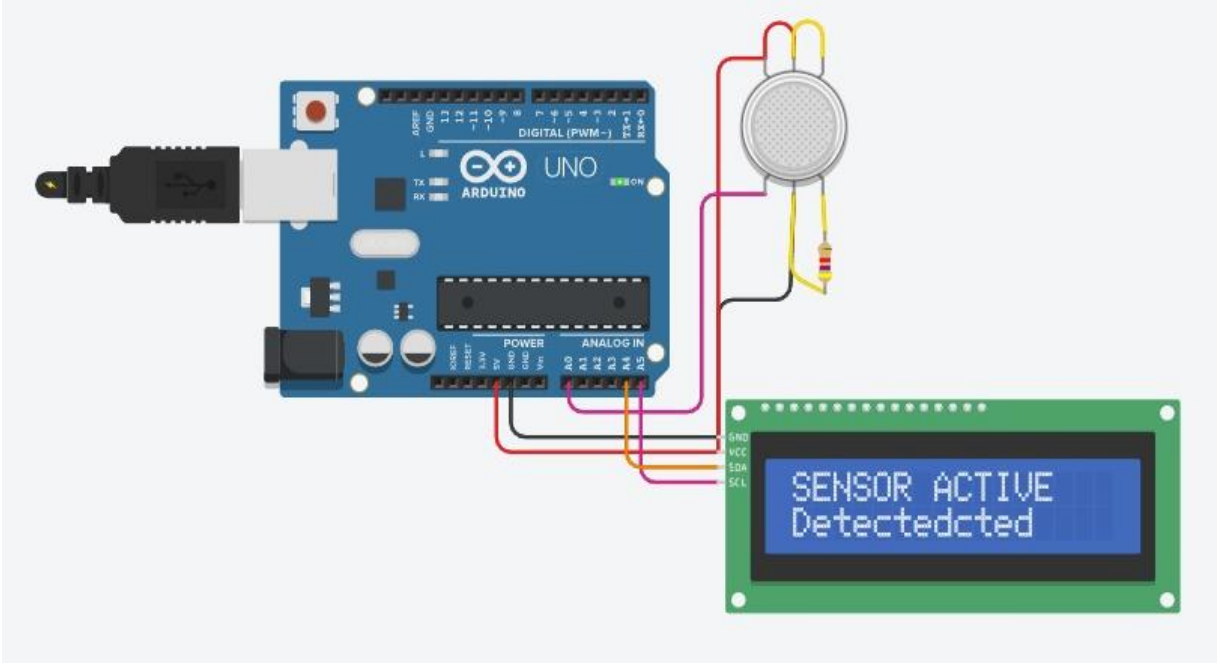
Experiment 8. Develop a program to detect the gas leakage in the surrounding environment.

AIM : Develop a program to detect the gas leakage in the surrounding environment.

COMPONENT:

Sl. NO	Name	Gas Sensor
1	Ardiuno Uno	**1**
2	Jumper Cable	**6**
3	Gas Sensor	**1**
4	MCP23008-based, 32 (0x20) LCD 16 x 2 (I2C)	**1**
5	Resistor (4.7 kΩ)	**1**

CIRCUIT DIAGRAM



SET UP :

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE



RESULT : Successfully demonstrated detect the gas leakage in the surrounding environment

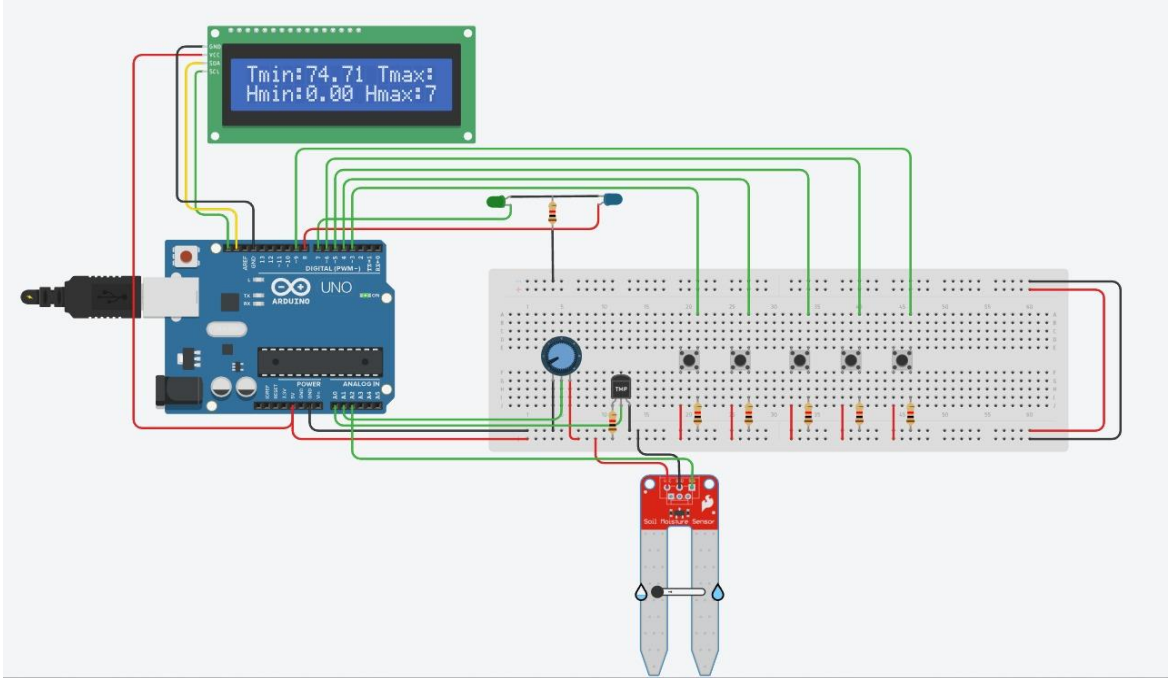
Experiment 9. Develop a program to demonstrate weather station readings using Arduino.

AIM : Develop a program to demonstrate weather station readings using Arduino.

COMPONENT:

Sl.NO	Name	Quantity
1	Ardiuno Uno	**1**
2	Jumper Cable	**10**
3	Bread Board	**1**
4	LED	**2**
5	Resistance (1KΩ)	**7**
6	LCD 16 x 2 (I2C)	**1**
7	Potentiometer (250 kΩ)	**1**
8	Temperature Sensor [TMP36]	**1**
9	Soil Moisture Sensor	**1**
10	Push Button	**5**

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
#include <Adafruit_LiquidCrystal.h>
#include <EEPROM.h>

Adafruit_LiquidCrystal lcd(0);
const int tempPin = A0;
const int humPin = A2;
const int baroPin = A1;

const int buttonPins[] = {3, 4, 5, 6,9};

const int ledPins[] = {7, 8};

float temperature, humidity, pressure, dewPoint;
float minTemp = 100, maxTemp = -100;
float minHumidity = 100, maxHumidity = 0;
float minPressure = 1000, maxPressure = 0;

float tempAlertThreshold = 30.0;
float humAlertThreshold = 80.0;
```

```
float tempHistory[10] = {0};
float humHistory[10] = {0};
float pressHistory[10] = {0};
int historyIndex = 0;
int currentScreen = 0;
const int eepromStartAddr = 0;
int eepromIndex = 0;

float calculateDewPoint(float temp, float hum) {
  float a = 17.27;
  float b = 237.7;
  float alpha = ((a * temp) / (b + temp)) + log(hum / 100.0);
  return (b * alpha) / (a - alpha);
}
void logDataToEEPROM(float temp, float hum, float pres) {
  int addr = eepromStartAddr + eepromIndex * 12;
  if (addr + 12 > EEPROM.length()) eepromIndex = 0;
  EEPROM.put(addr, temp);
  EEPROM.put(addr + 4, hum);
  EEPROM.put(addr + 8, pres);
  eepromIndex++;
}

void readSensors() {
  temperature = analogRead(tempPin) * 5.0 / 1024.0 * 100.0;
  humidity = map(analogRead(humPin), 0, 1023, 0, 100);
  pressure = analogRead(baroPin) / 10.0;
  dewPoint = calculateDewPoint(temperature, humidity);
```

```
    if (temperature < minTemp) minTemp = temperature;
    if (temperature > maxTemp) maxTemp = temperature;
    if (humidity < minHumidity) minHumidity = humidity;
    if (humidity > maxHumidity) maxHumidity = humidity;
    if (pressure < minPressure) minPressure = pressure;
    if (pressure > maxPressure) maxPressure = pressure;

    tempHistory[historyIndex] = temperature;
    humHistory[historyIndex] = humidity;
    pressHistory[historyIndex] = pressure;
    historyIndex = (historyIndex + 1) % 10;

    logDataToEEPROM(temperature, humidity, pressure);
}

void checkAlerts() {
    if (temperature > tempAlertThreshold) {
        digitalWrite(ledPins[0], HIGH);
    } else {
        digitalWrite(ledPins[0], LOW);
    }

    if (humidity > humAlertThreshold) {
        digitalWrite(ledPins[1], HIGH);
    } else {
        digitalWrite(ledPins[1], LOW);
    }
}
```

```
void exportData() {
  Serial.println("Exporting data:");
  for (int i = 0; i < eepromIndex; i++) {
    int addr = eepromStartAddr + i * 12;
    float temp, hum, pres;
    EEPROM.get(addr, temp);
    EEPROM.get(addr + 4, hum);
    EEPROM.get(addr + 8, pres);
    Serial.print("T:");
    Serial.print(temp);
    Serial.print(" H:");
    Serial.print(hum);
    Serial.print(" P:");
    Serial.println(pres);
  }
}

void displayGraph() {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("T:");
  for (int i = 0; i < 10; i++) {
  lcd.print(tempHistory[i] > tempAlertThreshold ? "*" : ".");
  }
  lcd.setCursor(0, 1);
  lcd.print("H:");
  for (int i = 0; i < 10; i++) {
    lcd.print(humHistory[i] > humAlertThreshold ? "*" : ".");
  }
```

```
}
String forecast = "N/A";

void calculateWeatherForecast() {
  float pressureChange = pressHistory[9] - pressHistory[6];
  float humidity = humHistory[9];
  float temperature = tempHistory[9];

  if (pressureChange < -2.0 && humidity > 70) {
    forecast = "Rain expected";
  } else if (pressureChange > 2.0 && humidity < 50) {
    forecast = "Clear skies";
  } else if (temperature > 30.0) {
    forecast = "Hot weather";
  } else if (temperature < 5.0) {
    forecast = "Cold, frost";
  } else {
    forecast = "Stable";
  }
}
void displayForecast() {
  lcd.clear();
 lcd.setCursor(0, 0);
  lcd.print("Forecast:");
  lcd.setCursor(0, 1);
  lcd.print(forecast);
}
void updateDisplay() {
  lcd.clear();
```

```
    if (currentScreen == 0) {
      lcd.setCursor(0, 0);
      lcd.print("T:");
      lcd.print(temperature);
      lcd.print("C H:");
      lcd.print(humidity);
      lcd.setCursor(0, 1);
      lcd.print("P:");
      lcd.print(pressure);
      lcd.print(" DP:");
      lcd.print(dewPoint);
    } else if (currentScreen == 1) {
      lcd.setCursor(0, 0);
      lcd.print("Tmin:");
      lcd.print(minTemp);
      lcd.print(" Tmax:");
      lcd.print(maxTemp);
      lcd.setCursor(0, 1);
      lcd.print("Hmin:");
      lcd.print(minHumidity);
      lcd.print(" Hmax:");
      lcd.print(maxHumidity);
    } else if (currentScreen == 2) {
      displayGraph();
    } else if (currentScreen == 3) {
      exportData();
      lcd.setCursor(0, 0);
      lcd.print("Export Complete");
    }
```

```
    else if (currentScreen == 4) {
        displayForecast();
    }
}

void handleButtonPress() {
    for (int i = 0; i < 5; i++) {
        if (digitalRead(buttonPins[i]) == HIGH) {
currentScreen = i;
            updateDisplay();
            delay(300);
        }
    }
}

void setup() {
    lcd.begin(16, 2);
    lcd.setBacklight(1);
    Serial.begin(9600);

    for (int i = 0; i < 4; i++) pinMode(buttonPins[i], INPUT_PULLUP);
    for (int i = 0; i < 2; i++) pinMode(ledPins[i], OUTPUT);

    lcd.setCursor(0, 0);
    lcd.print("Weather Station");
    delay(2000);
    lcd.clear();
}
```

```
void loop() {
    readSensors();
    checkAlerts();
    updateDisplay();
    handleButtonPress();
    calculateWeatherForecast();

    delay(2000);
}
```

RESULT : Successfully demonstrated weather station readings using Arduino

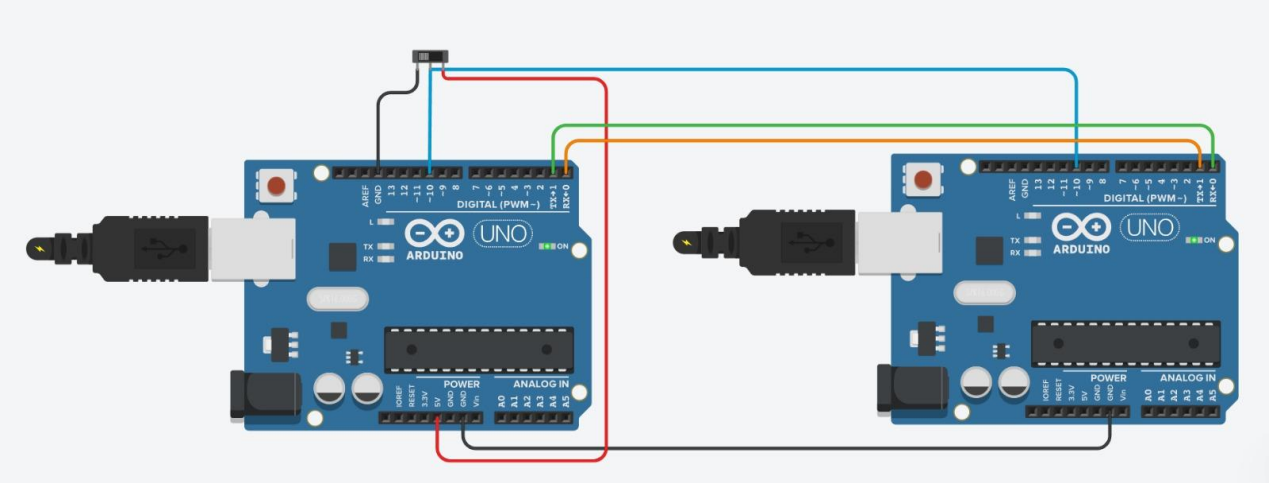
Experiment 10. Develop a program to setup a UART protocol and pass a string through the protocol.

AIM: Develop a program to setup a UART protocol and pass a string through the Protocal.

COMPONENT:

Sl.NO	Name	Quantity
1	Ardiuno Uno	**2**
2	Jumper Cable	**7**
3	Slideswitch	**1**

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

f) Then goto code serial monitor to check working output

CODE

```
const int MAX_LEN =30;
char sendMsg[MAX_LEN] = "Hello i'm Arduino-2\n";
char receiveMsg[MAX_LEN];
int switchState = 1; // Active HIGH input
int switch_pin = 10;

void setup() {
  Serial.begin(9600);
  pinMode(switch_pin, INPUT);
}

void loop() {
  switchState = digitalRead(switch_pin);
  if (switchState == HIGH) {
    // Receive if switch is off
    if (Serial.available() > 0) {
      int len = Serial.parseInt();        // Read the length as number
      Serial.read();                      // Consume the newline after the number

      int n = Serial.readBytes(receiveMsg, len);
      receiveMsg[n] = '\0';               // Null-terminate the received string

      Serial.print("Message = ");
      Serial.println(receiveMsg);
    }
  } else {
   // Send if switch is on
    int len = strlen(sendMsg);
    Serial.write(receiveMsg, len); // send raw message
   delay(1000);
  }
}
```

RESULT : Successfully demonstrated UART using two Arduino

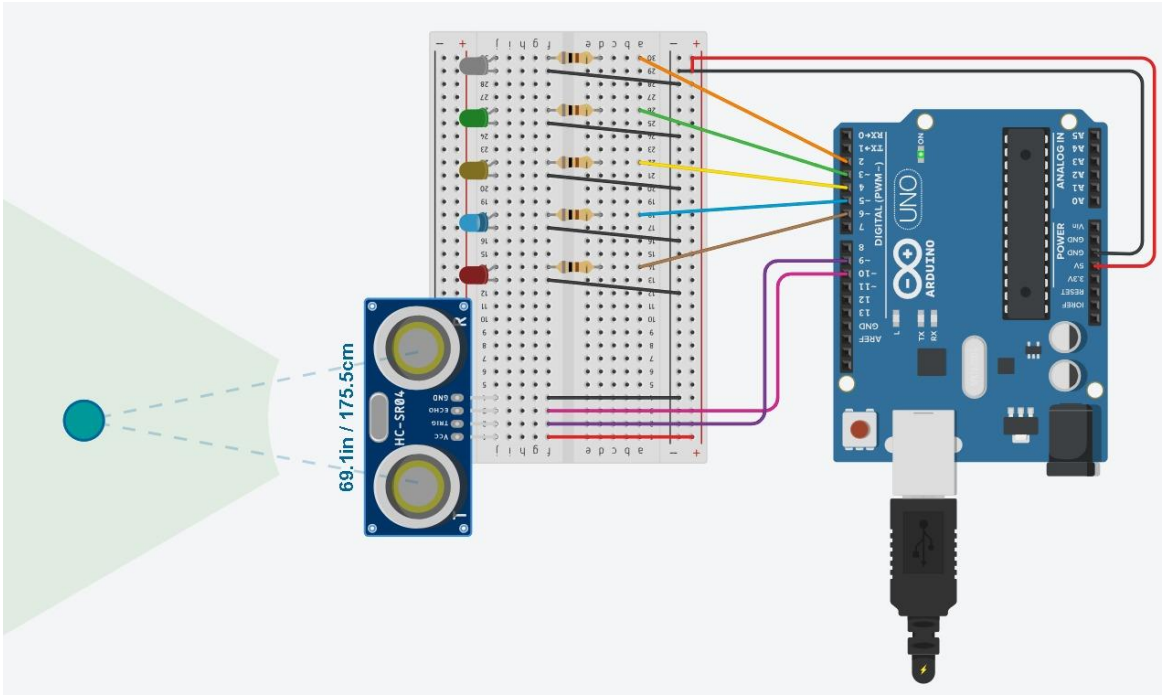
Experiment 11. Develop a water level depth detection system using Ultrasonic sensor.

AIM : Develop a water level depth detection system using Ultrasonic sensor.

COMPONENT:

Sl.NO	Name	Quantity
1	Ardiuno Uno	1
2	Jumper Cable	17
3	Bread Board	1
4	LED	5
5	Resistance (800 Ω)	5
6	Ultrasonic Distance Sensor (4-pin)	1

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.
b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.
c) Open Arduino IDE then goto tools and select appropriate Arduino board.
d) Select tool then select the port select the com port to which board is connected.
e) Type sketch (Program) and upload to board.

CODE

```
 int trigPin=9;
int echoPin=10;
int a=2,b=3,c=5,d=6,e=4;
long dist;
long ival;
void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(a,OUTPUT);
  pinMode(b,OUTPUT);
  pinMode(e,OUTPUT);
  pinMode(c,OUTPUT);
  pinMode(d,OUTPUT);
  Serial.begin(9600);
}
void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  ival=pulseIn(echoPin,HIGH);
  dist=(ival/2)/29.154;
  Serial.print("dist:");
  Serial.print(dist);
  Serial.println("CM");
  if(dist<=50){
  digitalWrite(a,HIGH);
    digitalWrite(b,LOW);
    digitalWrite(e,LOW);
    digitalWrite(c,LOW);
    digitalWrite(d,LOW);
  }
    else if(dist<=100){
      digitalWrite(a,LOW);
```

```
    digitalWrite(b,HIGH);
    digitalWrite(e,LOW);
    digitalWrite(c,LOW);
    digitalWrite(d,LOW);
    }
    else if(dist<=150){
    digitalWrite(a,LOW);
    digitalWrite(b,LOW);
    digitalWrite(e,HIGH);
    digitalWrite(c,LOW);
    digitalWrite(d,LOW);
    }
    else if(dist<=200){
      digitalWrite(a,LOW);
    digitalWrite(b,LOW);
    digitalWrite(e,LOW);
    digitalWrite(c,HIGH);
    digitalWrite(d,LOW);
    }
    else if(dist<=250){
      digitalWrite(a,LOW);
    digitalWrite(b,LOW);
    digitalWrite(e,LOW);
    digitalWrite(c,LOW);
    digitalWrite(d,HIGH);
      }
    else
    {
     digitalWrite(a,HIGH);
    digitalWrite(b,HIGH);
    digitalWrite(e,HIGH);
    digitalWrite(c,HIGH);
    digitalWrite(d,HIGH);
  }
delay(50);
}
```

RESULT: Successfully demonstrated water level depth detection system using Ultrasonic sensor

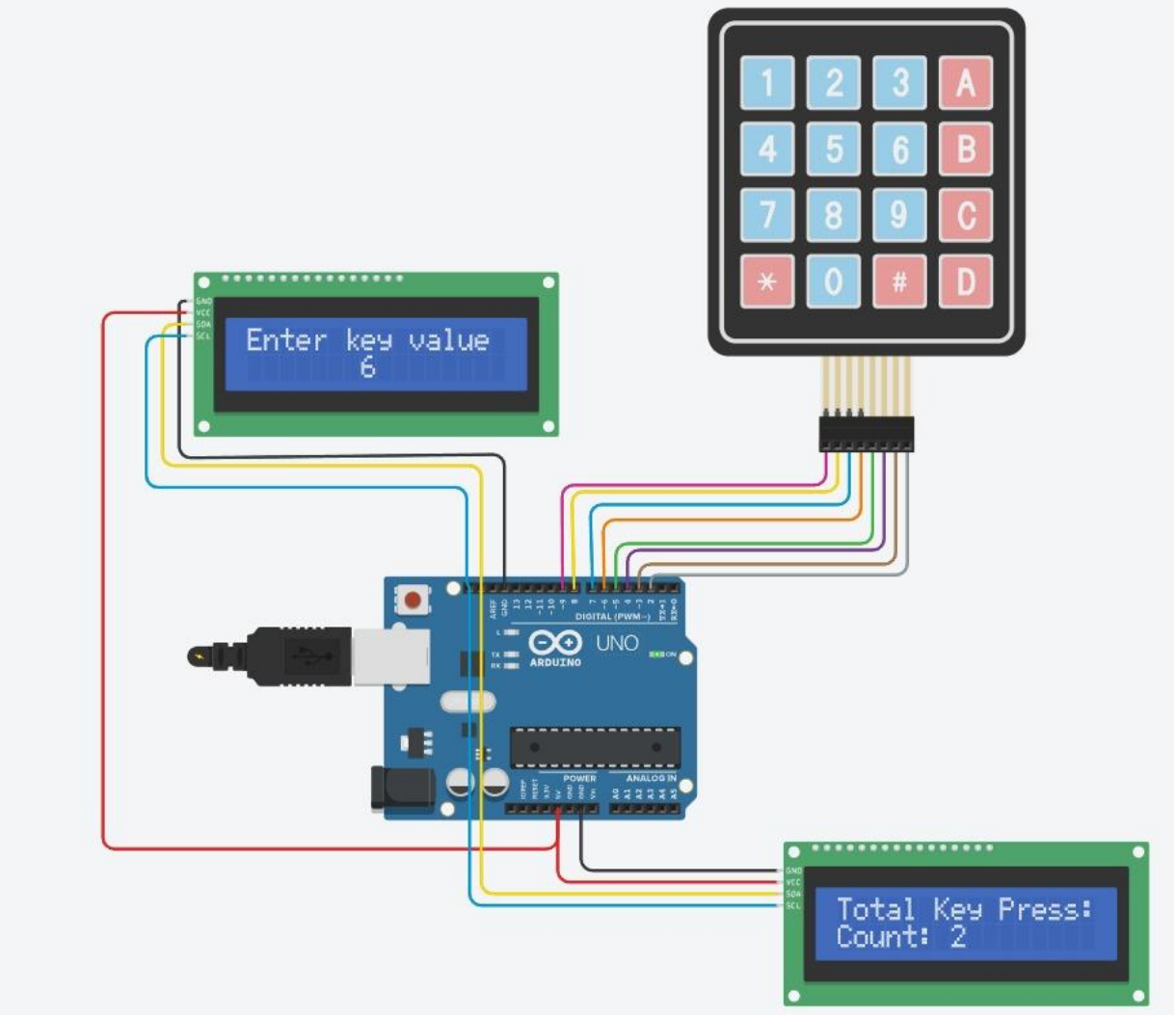
Experiment 12. Develop a program to simulate interfacing with the keypad module to record the keystrokes.

AIM: Develop a program to simulate interfacing with the keypad module to record the Keystrokes.

COMPONENT:

Sl.no	Name	Quantity
1	Ardiuno Uno	**1**
2	Jumper Cable	**8**
3	Bread Board	**1**
4	Keypad 4x4	**1**
5	LCD 16 x 2 (I2C)	**2**

CIRCUIT DIAGRAM:



SET UP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then goto tools and select appropriate Arduino board.

d) Select tool then select the port select the com port to which board is connected.

e) Type sketch (Program) and upload to board.

CODE

```
 #include <Adafruit_LiquidCrystal.h>
#include <Keypad.h>

// LCD 1 – Shows key pressed
Adafruit_LiquidCrystal lcd_1(0x20);

// LCD 2 – Shows total key count
Adafruit_LiquidCrystal lcd_2(0x21);

const byte ROWS = 4;
const byte COLS = 4;

char hexaKeys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'},
};

byte rowPins[ROWS] = {9, 8, 7, 6};
byte colPins[COLS] = {5, 4, 3, 2};

Keypad customKeypad = Keypad(makeKeymap(hexaKeys), rowPins,
colPins, ROWS, COLS);

int keyPressCount = 0;
void setup() {
  Serial.begin(9600);
```

```
  // Initialize both LCDs
  lcd_1.begin(16, 2);
  lcd_1.setBacklight(1);
  lcd_1.print("Enter key value");
  lcd_1.setCursor(0, 1);

  lcd_2.begin(16, 2);
  lcd_2.setBacklight(1);
  lcd_2.print("Total Key Press:");
  lcd_2.setCursor(0, 1);
  lcd_2.print("Count: 0");
}
void loop() {
  char customKey = customKeypad.getKey();
  if (customKey) {
    Serial.println(customKey);

    // Display key on LCD 1
    lcd_1.setCursor(7, 1);
    lcd_1.print(" ");
    lcd_1.setCursor(7, 1);
    lcd_1.print(customKey);

    // Increment and show count on LCD 2
    keyPressCount++;
    lcd_2.setCursor(0, 1);
    lcd_2.print("Count: ");
    lcd_2.print(keyPressCount);
    lcd_2.print("   "); // Clear extra digits if count goes down

    delay(100);
  }
}
```

RESULT: Successfully demonstrated simulate interfacing with the keypad module to record the keystrokes.